Command Broker Framework

Cross-Sector Human Interposition Requirements

Applicable Across Critical Infrastructure and Democratic Governance Domains

Companion Framework to the DBA Sector Series and the Democratic Safeguards Project

Version 0.5 — DRAFT

May 2025 · Revised November 2026

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For review by sector regulators, AI governance bodies, and critical infrastructure policy stakeholders

1. Purpose and Scope

This document establishes a cross-sector framework for the Command Broker function — the human interposition requirement applicable wherever the output of a component within scope can produce a consequential action in a critical process, physical or otherwise. Scope is the complement of the determinism carve-out and is governed by FD-LD §4.3; Generative AI (Gen AI) systems are its principal case and not its definition. It defines the broker concept, the conditions under which it applies, the architectural requirements that give it force, and the qualification criteria that determine who may fill the role.

The framework is organized at the cross-sector level because the broker requirement is not sector-specific. It derives from a sector-independent property — behavior not shown to be bounded by proof — of which the non-determinism of Gen AI systems is the principal case (FD-LD §4.3). The same prompt can produce different outputs across execution cycles. Formal verification methods that bound the behavior of deterministic systems do not apply. In any domain where the output of an in-scope component can initiate or materially influence a consequential action — whether that action is a change to a physical process parameter or the publication of content intended to influence political opinion — the same functional requirement applies: a qualified human must be interposed between that output and the consequential action, and that interposition must be enforced by architecture, not by procedure or operator discipline alone.

Two applications of this requirement are currently being developed within the Eclectic Technologies body of work. The Command Broker governs the deployment of in-scope components, Gen AI principal among them, in critical infrastructure industrial control systems, where the consequential action is a change to a physical process parameter. The Influence Broker governs Gen AI deployment in the information environment, where the consequential action is the publication or amplification of content capable of influencing political or social outcomes at scale. Both are developed in this framework. Both are derived from the same source requirement: the non-determinism of Gen AI and the consequential nature of its outputs.

Sector-specific implementation details — consequence thresholds, band assignments, and qualification standards adapted to the sector's operational context — are set out in sector annexes to this framework. The annexes currently in development or anticipated are: DBA-W (water sector), DBA-GAS (natural gas transmission), DBA-OIL (petroleum sector), DBA-Nuclear (nuclear facilities), and the Democratic Safeguards project (information environment and democratic governance). The cross-sector framework establishes the architecture and qualification philosophy; the annexes apply them.

2. The Broker Concept — A General Principle

The broker is a human decision-maker who stands between an AI system's recommendation and the consequential action it would initiate or influence. The broker's function is not to relay the AI's recommendation. It is to evaluate the recommendation against independent operational knowledge, current conditions, and professional judgment, and to authorize, modify, or reject the recommended action before it takes effect. The broker exercises genuine decision authority. The interposition is substantive, not ceremonial.

The broker concept is necessary because of a specific property of Gen AI that distinguishes it from the machine learning (ML) systems that preceded it in industrial and information environments. ML systems — anomaly detection, predictive maintenance, pattern recognition algorithms — are deterministic within bounded input ranges. Given bounded inputs, they produce bounded, testable outputs. Their behavior can be characterized through semi-formal methods. Where such a component is deterministic within a stated input bound and formally testable in the deployed configuration, it meets the carve-out and is outside the Command Broker obligation (FD-LD §4.3). The carve-out removes that obligation only; it is not a grant of ungoverned autonomy, and it is not a finding that the component's actions are below the line.

Gen AI systems, including large language models and related architectures, are not deterministic in this sense. The same input can produce meaningfully different outputs across execution cycles. The statistical processes that generate Gen AI outputs do not lend themselves to formal verification by methods that bound ML behavior. This is not a temporary limitation of current architectures that better engineering will resolve. It is a structural property of systems that generate novel outputs from learned distributions. That property does not make Gen AI unsuitable for deployment in consequential domains — but it does make unmediated autonomous action by Gen AI systems unsuitable wherever the consequences of error are significant.

The broker requirement is the response to that structural property. It is derived from the author's analysis of ML and Gen AI risk in critical infrastructure industrial control systems, refined through the ISAGCA Supplier Ad-hoc Working Group, which he convened and moderated from its first working session in February 2024. ISA published a snapshot of that analysis (Version 20) in 2025; the work has continued independently since and appears in its current edition as "Generative AI Risk in Industrial Control Systems" (Version 29, Eclectic Technologies, 2026). That work establishes a three-tier criticality framework applicable across all critical infrastructure sectors. This framework extends the broker concept beyond the industrial control system domain to encompass the information environment, where Gen AI's non-determinism poses a structurally equivalent risk.

Mechanism and accountability. The broker requirement has two parts that must not be conflated. The first is the enforcement mechanism — the means by which a consequential action is held until the interposition is satisfied. Where a consequence reduces to a checkable physical or operational constraint, that mechanism is implementation-neutral: an interlock, a hard limit, or

a verified deterministic component may perform the enforcement, and does so more reliably than operator discipline. The second is the accountable authority — the party that holds the duty of care, answers to a reviewer, a regulator, or a board of inquiry for the decision, and whose judgment the audit record exists to make defensible. Accountability is not a property that added constraints confer; it requires an entity that can bear an obligation and be answerable for an outcome. A non-deterministic AI system cannot be that entity, and neither can a deterministic mechanism.

It follows that the question of whether a sufficiently constrained AI system could satisfy the broker criteria is answered by this distinction rather than by a prohibition. An AI system, or an automated component, may implement the enforcement mechanism where the consequence is checkable; it may not be the broker, because the broker is the accountable authority, and that authority remains with a named human decision-maker or governance body. Where a consequence cannot be reduced to a deterministic check — an integrity failure, an influence effect, or an accurate recommendation framed to induce an unsafe decision — accountability is non-delegable, and the interposition must be discharged by a human broker whose reasoning is preserved in the decision record. This principle applies equally to the Command Broker and the Influence Broker defined below.

3. The Command Broker

3.1 Definition

The Command Broker is the human operator who decides — who brokers — a course of physical action based on process observables and output from a component within scope of the Bright Line. A component is within scope when it does not meet the determinism carve-out. FD-LD §4.3 states that scope rule and governs, and it is cited here rather than restated; generative components are the principal case within scope and not its definition. The Command Broker function exists to ensure that no output of an in-scope component makes an above-line action — one action as FD-BL §4.3 defines it, placed on its consequence under FD-BL §4.2 — without the Command Broker’s authorization, discharged in the moment or, where FD-BL-D1 applies, in advance as the author of the pre-authorized envelope (FD-BL §4.5; FD-BR §4.2). A component is carved out by the property test FD-LD §4.3 sets and not because it is machine learning, and the carve-out removes it from this obligation only: it is not a grant of ungoverned autonomy, and it is not a finding that the component’s actions are below the line.

The Command Broker is not a procedural role that can be satisfied by policy language, training certification, or workflow documentation alone. The Command Broker architecture requires that the control system enforce the broker function mechanically: the system must be incapable of executing an above-line physical process command originating from a component within scope (FD-LD §4.3) without the required authorization. In the moment, that authorization is a positive human confirmation step that is local, auditable, and not itself executable remotely or by another

software component. In advance, it is the Command Broker’s authorization of a pre-authorized envelope and acceptance of its completeness proof, after which a verified gatekeeper satisfying FD-BL-D1 R1–R3 admits each action in the class: the human discharges, and the gatekeeper admits (FD-BR §4.1). No other form of pre-authorization discharges the obligation.

3.2 The Bright Line

The Bright Line is defined in FD-BL — Foundational Definition — The Bright Line, which governs. It is a threshold on actions, not a boundary between architectural layers: an action is above the line when its incorrect performance could produce a consequence at or above the facility's severity threshold (FD-BL §4.1–§4.2). An above-line action may not execute without the Command Broker’s authorization, given in the moment or, where FD-BL-D1 applies, in advance (FD-BL §4.5), and that requirement is enforced by architecture, not by procedure. The advisory-to-command seam is a useful place to build that enforcement; it is not the definition of the line, and a single gate placed there on the assumption that it coincides with the line will leave above-line actions ungated on the advisory side and gate below-line actions needlessly on the command side (FD-BL §4.2). A vendor's characterization of embedded generative AI as “autonomous process optimization” or “direct control integration” is a product description, not a compliance determination; placement follows what the component's actions can cause. The Design Basis requirement governs; the vendor characterization does not.

The Bright Line is based on consequence analysis, not on system sophistication. The relevant question is not how capable the Gen AI system is, but what happens if it produces an erroneous output that is acted upon. A simple AI model that recommends a chemical dosing adjustment at a water treatment plant is above the Bright Line because its erroneous output could lead to a public health consequence. A sophisticated AI system performing energy optimization in a data center may be below the Bright Line if the consequence of its erroneous output does not reach the defined consequence threshold. Sophistication does not govern. Consequence governs.

§3.2a Placement is determined action by action. The unit of placement is the individual action, not the system or the application. A single generative component deployed once in one facility will typically produce actions on both sides of the line, and output that makes no change of its own. Summarizing historian data makes no change of its own: it is placed below the line as a process action and carries the indication-integrity obligation of FD-BL-D2 (FD-BL §4.2). Drafting a maintenance work order that, once acted upon, would take equipment out of service is an action placed on that consequence. Recommending a setpoint change during operation is placed on what an incorrect change could cause, and the recommendation and its acceptance are one action (FD-BL §4.3), which also defines what one action is. The required architecture is therefore a consequence classifier that routes action by action, not a single gate at a seam; the classifier's routing decisions and their basis are part of the evidentiary record.

Placement is distinct from criticality-band assignment (§3.3). Band assignment is a property of the application, derived from its consequence-severity profile, and sets the rigour of assessment,

qualification, and audit. Placement is a property of each action and sets whether that action may proceed without human authorization. A Highest-band application contains below-line actions; a Lowest-band application may contain above-line ones. Neither determines the other.

3.3 Criticality Bands

Applications are assigned to one of three criticality bands — Highest, Mid, Lowest — by the severity of the worst credible consequence of a cognitive error, measured across the six consequence categories and aggregated by taking the maximum (see the Consequence-Severity Tier Model, which is the normative basis for this assignment, and §5.3 below). The band for any specific application is fixed by consequence analysis in the applicable sector annex. The bands are:

Highest Criticality. The most rigorous assessment, qualification, and audit regime applies. Whether any specific action within such an application may proceed autonomously is decided by its placement under FD-BL §4.2 (above-line actions require a Command Broker; below-line actions do not), not by whether the producing component is ML or Gen AI. Determinism decides only whether a component is in scope of Bright Line analysis (FD-LD §4.3).

Mid-Level Criticality. A significant but bounded and recoverable worst credible consequence; assessment, qualification and audit are proportionate to that severity. Whether any specific action within such an application may proceed autonomously is decided by its placement under FD-BL §4.2, exactly as at the Highest band, and not by the band. Where the band is entered through maloperation — where incorrect performance could produce the consequence that sets it — an above-line action necessarily exists, and a prohibition on fully autonomous operation of an in-scope component follows from placement. Where it is entered through loss of service or availability rather than through wrong action, no above-line action is guaranteed and any such prohibition is a conservative default rather than a consequence of placement. An instrument relying on this band states which of the two it means.

Lowest Criticality. No reach to a critical function, to life-safety, or to a critical service; the lightest assessment, qualification and audit regime applies. A Lowest-band application may nonetheless contain above-line actions, and any such action requires a Command Broker on its placement. The band lightens the rigour of assessment; it does not license autonomy.

Naming crosswalk (legacy reading aid). The band names used here are canonical. Earlier documents in the corpus, and the ISA Parent Framework, numbered the same bands Tier 1 / 2 / 3; that numbering is retained below only to read older text. The parenthetical type-words that once accompanied the numbers — safety-critical, quality-of-service, administrative — are not band definitions; they echo the three tiers of the error axis, which the Tiered Assessment Framework defines as safety-critical, quality-of-service and compliance-driven, each assessed at any band. New work uses the band names.

Criticality band (canonical)	Legacy number	Severity — worst credible

		consequence, any category
Highest	Tier 1	Design-Basis level: population- or mission-scale, loss of a critical function, or life-safety
Mid	Tier 2	Significant but bounded and recoverable, below the Design-Basis threshold
Lowest	Tier 3	No reach to a critical function, life-safety, or critical service in any category

3.4 Command Broker Qualification

The qualification criteria for a Command Broker must be derived from the operational context of the sector and facility in which the broker function is performed. The cross-sector framework establishes the qualification methodology; sector annexes define the specific criteria. The methodology requires that the following dimensions be addressed in each annex:

- Technical competency. The Command Broker must possess sufficient technical knowledge of the physical process to evaluate an AI recommendation independently — not to replicate the AI's analysis, but to recognize when the recommendation is inconsistent with current conditions, operating constraints, or applicable limits. A Command Broker who accepts AI recommendations without independent evaluation is not performing the broker function.
- Process authority. The Command Broker must hold formal authority to authorize or reject changes to the relevant process parameters. The broker role cannot be assigned to an observer, an advisor, or a position that requires escalation before acting. The broker decides.
- Situational awareness. The Command Broker must have real-time access to the process observables needed to evaluate AI recommendations in context. A broker who can see only the AI's recommendation, without the underlying process data the recommendation is based on, cannot perform an independent evaluation.
- Adversarial stress competency. The Command Broker must be qualified to perform the broker function under degraded conditions — loss of normal communications, partial loss of sensor feeds, elevated operational tempo, or conditions arising from a military action or cyberattack scenario. Broker qualification programs must include training and evaluation under degraded conditions, not only under normal operating assumptions. This dimension is addressed in detail in Section 3.6.

3.5 Audit and Verification

Architectural enforcement of the broker function is necessary but not sufficient. The Command Broker must demonstrably be in the loop, not nominally so. Audit requirements must address three questions:

- Is the broker confirmation step being performed before physical commands execute, or is it being bypassed through workarounds, pre-authorization outside FD-BL-D1 R1–R3, or system configuration that effectively bypasses the enforcement mechanism? Where the authorization is discharged in advance, the record shall show the named Command Broker's authorization of the envelope and acceptance of its proof, the classifier's routing basis, and the gatekeeper's admission of each action (FD-BL-D1 §5, §8).
- Is the broker's confirmation substantive — reflecting an independent evaluation of the AI recommendation — or is it a pro forma acknowledgment? Audit records must capture sufficient information to distinguish between genuine evaluation and rubber-stamping.
- Is the broker qualification maintained current? An individual qualified at the time of system deployment who has not maintained technical competency as the process or the AI system has changed is not a qualified broker. Qualification must be maintained, not merely achieved.

3.6 Command Broker Function Under Stress Conditions

The Design Basis Accident — Military Action (DBA-MA) framework and the DBA sector series establish that the Design Basis for critical infrastructure must account for conditions more severe than normal operation, including deliberate adversarial action. The Command Broker qualification framework must reflect this. A broker qualified only for normal operating conditions is under-qualified for the environments in which the broker's function is most consequential.

Stress conditions relevant to Command Broker qualification include: loss of external communications requiring the broker to act on local information alone; partial degradation of sensor feeds requiring the broker to identify and compensate for missing or unreliable data; elevated operational tempo in which multiple recommendations require evaluation in rapid sequence; and conditions in which the AI system itself may be operating from compromised or manipulated inputs — the cyberattack scenario in which the AI's recommendation is itself adversarially influenced.

The qualification program for Command Brokers at the Highest-band facilities must include scenario-based evaluation for each of these stress categories. Sector annexes specify the scenario content appropriate to their operational environments.

3.7 The Broker's Interface — What the Broker Must Be Shown

The broker function is discharged through an interface, and an interface that withholds what the decision needs converts a substantive authorization into a ceremonial one. Section 3.4 requires that the broker have real-time access to the process observables a recommendation rests on;

Section 3.5 requires that the confirmation be substantive rather than pro forma. Those two requirements meet at the interface, and this section states what the interface must carry for them to be satisfiable. What each resolves to in a particular control room is the sector annex's to state.

The mode this section governs. The requirements below are stated for the attended case: a qualified broker is present, a recommendation is before him, and the interface must carry what the decision needs. That is not the only mode in which the authorization required at FD-BL §4.5 is discharged. Under FD-BL-D1 the authorization may be discharged in advance, over a pre-authorized envelope whose completeness proof the Command Broker, discharging in advance, has accepted, where in-the-moment authorization is infeasible; and a sector annex may specify bounded conditions under which no broker is at the interface at all. The advance-discharge mode does not escape these obligations. It relocates them, to the Command Broker discharging in advance, who defines the envelope and accepts its proof (FD-BR §4.2; FD-BL-D1 §7), and who needs the same three things translated: the calibration and out-of-distribution status of whatever component proposes into the gatekeeper, a consequence statement for the envelope's bounds rather than for a single action, and a display of whether the envelope is in force, stale, or locked. What those obligations require in a given sector, and what the display obligation converts into where there is no live placement to display, are the sector annex's to state, and each annex shall state them. This framework does not leave the advance-discharge mode without interface obligations by stating the attended case alone.

The conditions these requirements compensate, and the criterion of adequacy that follows. The three requirements below are not general good practice at an interface. They are compensating requirements, written against three conditions that degrade human decision-making at exactly the moment the broker function matters most: cognitive overload, automation bias amplified by stress, and time compression. Under the conditions the DBA-MA envelope contemplates, the three are present simultaneously and at maximum intensity. This framework nonetheless states the requirements as unconditional standing obligations that bind in normal operation and not only under duress, and that enlargement is deliberate rather than an omission of the trigger: an interface built to be adequate only when the control room is calm is not adequate, and a requirement that is honoured only under stress is a requirement that will not have been built. What the three conditions supply, and what the requirements cannot be read without, is the criterion of adequacy. Each obligation below is discharged only in a form that remains effective under cognitive overload, under stress-amplified automation bias, and under time compression. The criterion is not decorative. A consequence preview that is complete, accurate and four screens long satisfies the words of the requirement and fails entirely under time compression; a calibrated confidence rendered as a distribution the broker must interpret satisfies the words and adds to cognitive load rather than relieving it; a placement display that is continuous, correct and visually quiet satisfies the words and does nothing against automation bias, which is the condition under which a quiet indication is precisely what is not questioned. An interface can satisfy each of the three requirements in a form that defeats the condition each

exists to compensate, and the sector annex's qualification, assessment and audit criteria are where that is caught.

Calibrated confidence, with its basis and its uncertainty stated. A recommendation presented without an indication of how well founded it is asks the broker to supply a judgment the proposing component already holds and did not share. The interface must present the proposing component's confidence in a calibrated form — one whose reported values have been validated against observed outcomes — together with the basis on which the recommendation rests and an explicit statement of what is not known. Uncalibrated confidence is worse than none, because a number that does not track outcomes invites reliance it has not earned. Where the proposing component is operating outside the conditions its training distribution covers, the interface must say so rather than report a confidence computed as though it were not. Disclosure begins that treatment and does not complete it: a confidence disclosed as computed outside the component's training distribution is not a confidence on which an above-line authorization may rest, and the authorization stands or falls on the independent evaluation the sector annex requires. A sector whose above-line decisions are taken in a regime that is by construction absent from any operating history will find this the ordinary case and not the exception, which is why the annex and not this section states what that independent evaluation is.

Consequence preview. The interface must state what the recommended action is expected to do: its expected effect on the process, the observables that would confirm that effect, and the consequence category against which the action's placement was decided. A broker authorizing an above-line action is authorizing its consequence, and cannot do so substantively while the consequence is implicit. The preview is not a prediction of success. It is a statement of what the action is for and of what would show that it worked, which is what allows a wrong recommendation to be recognized before it executes rather than after. The category named is the one against which placement was decided, which is not necessarily the category a facility's own event classification would assign to the same action, and the requirement is to the former. The observables the preview nominates are presented quantities within the meaning of the integrity obligation stated below: a confirmatory observable offered without its integrity status hands the broker a confirmation path that can be forged.

Continuous placement-state display. The interface must show, continuously and without the broker having to ask, whether the action under consideration is above or below the Bright Line and whether that placement is live or stale. Placement is decided action by action (§3.2a) and can change with plant state, so a display computed under a state that no longer holds tells the broker something worse than nothing: it tells him something false with the authority of the system behind it. Three placement states must be distinguishable — above the line, below the line, and placement indeterminate — and an indeterminate placement is treated as above the line until it is resolved. The display is keyed to the action and not to the component. The same component may be outside the Command Broker obligation for one action and above the line for the next, and a component-level indicator cannot show that; the requirement's own unit — the action under

consideration — already says at what grain the display is built, and it is said plainly here because an implementer reading quickly will build the indicator at the wrong one.

The third requirement has a plant-side precondition, and the first two do not. The first two requirements are obligations on the interface and not on the plant, and of them that is true without qualification. It is not true of the third. A continuous, live-or-stale placement display cannot be produced by the interface by itself: it needs a current, authenticated statement of plant condition, delivered over a path that survives the conditions under which the display matters most, and under denial that is the obligation most likely to fail. Where the interface cannot obtain an authenticated current state it shows the placement as indeterminate, which by the rule above is treated as above the line until it is resolved. Stated that way the boundary holds and the default does the work. What that path is, what it must survive, and what authenticates the state it carries belong to the sector annex and to the communications design basis, and not to this section.

A carve-out is scope, and not a placement. Three placement states are not the whole of what the display must distinguish, because this framework has a fourth condition that is not a placement at all. A component that meets the carve-out — deterministic within a stated input bound and formally testable in the deployed configuration — is outside the Command Broker obligation (FD-LD §4.3; §3.1). It is not below the line: a deterministic protective function may take some of the highest-consequence actions in a facility, and rendering its actions as below the line would be false. The display shall therefore never render an action carved out of the obligation as below the line. Where the carve-out applies it is shown as scope — the action is outside the obligation — and no placement is asserted for it. An express rule is preferred here to a fourth state because the carve-out removes an obligation and does not decide a placement, and a fourth state would invite the reader to treat it as a third place on the line. This is the same correction the Tiered Assessment Framework carries at §2.4 for the scope of the obligation itself — the carve-out is not the line — and the two are one correction made in two places. Stating it at the display is what keeps the error from being taught back, continuously and with the system's authority, to every operator who watches the surface.

What is presented must carry its integrity status. Each of the three requirements above is an obligation to present, and an obligation to present is not an obligation to authenticate. An interface that has just been required to carry three new decision-bearing quantities has been given three new channels, each authoritative to the broker by construction, and where the adversary's object is the decision path rather than the plant, that is where the targeting lands. A broker who would have hesitated in front of a bare recommendation does not hesitate in front of one carrying a calibrated confidence, a consequence preview and a live green placement state, which is the reason to forge them. The three presented quantities therefore carry the indication-integrity obligation of FD-BL-D2, which governs and is cited rather than restated: output of an in-scope component that can function as the basis for an above-line authorization, or as an input on which placement or discharge depends, shall be traceable to independently verifiable values and shall not be the sole basis for that authorization. Two consequences are stated here because they are

interface consequences. An integrity status that cannot be established is itself displayed rather than omitted, so that a quantity whose trustworthiness is unknown is shown as unknown and not shown plain. And where the basis is not trustworthy, the above-line authorization is not available on that basis: the placement resolves to indeterminate and, by the rule above, is treated as above the line. This framework states that floor rather than leaving the failure posture unstated, because an interface obligation with no stated consequence for its own non-satisfaction is discharged by degradation — the display goes grey and the authorization proceeds. What a sector requires of the broker once the basis has failed, including any conservative fallback to conventional instrumentation and established operating procedure, is the sector annex's to state, and each annex shall state it.

These are cross-sector interface obligations; their sector instantiations are not. What a consequence preview previews against when the consequence is a design-basis one, what the state display shows in a mode where an action's placement changes, and what calibration means when the proposing component's training distribution excludes the plant's accident space are questions each annex answers for its own domain. The requirements originate in the retirement of DBA-AI-CBR and were answered at D-1 on 4 September 2026 as cross-sector interface obligations with the nuclear instantiation in CB-Annex-Nuclear; the text is composed here rather than moved, no source text for it having existed.

4. The Influence Broker

4.1 Definition

The Influence Broker is the human decision-maker who stands between a Gen AI system’s content output and its publication or amplification to an audience that can be influenced at scale. The Influence Broker’s function is to evaluate AI-generated content for accuracy, contextual appropriateness, and potential to distort political or social outcomes, and to authorize, modify, or reject it before it reaches an audience.

The Influence Broker applies in any deployment context where Gen AI is used to generate content — text, image, audio, or synthetic media — that is intended to inform, persuade, or mobilize an audience, and where that audience is sufficiently large or strategically positioned that the content’s effects can be consequential at a societal level. The Influence Broker concept was developed in the Democratic Safeguards project. This section establishes its relationship to the Command Broker and its common derivation from the broker principle.

4.2 The Common Derivation

The Command Broker and the Influence Broker are the same functional requirement applied to different consequence domains. The structural parallel is precise.

In the Command Broker context, a Gen AI system produces a recommendation; if acted upon without review, the recommendation could change a physical process parameter, potentially causing physical harm. The broker interposes between the recommendation and the physical action.

In the Influence Broker context, a Gen AI system produces content; if published without review, that content would enter the information environment, potentially distorting political or social outcomes. The broker interposes between the content and its publication.

The parallel holds at the level of mechanism, not merely at the level of analogy. In both cases, the Gen AI system is non-deterministic and cannot be formally verified; the output is consequential; the consequence cannot be recalled once the action is taken; and the only reliable protection against erroneous or manipulated output producing a consequential result is a qualified human evaluation before the action is completed. The Bright Line concept applies equally, and it applies as §3.2 states it: a threshold on actions, placed on the consequence of the action, and not the boundary between the advisory layer and the consequential action. The advisory-to-command seam is where interposition is built in the industrial domain and the publication seam is its counterpart in the information environment; in neither case is the seam the line.

4.3 On the Question of Politicization

The inclusion of the Influence Broker in this framework alongside the Command Broker will, for some readers, raise the question of whether a document that addresses both industrial control system requirements and democratic governance requirements has become a political document. That concern is understandable, and it is worth addressing directly.

This framework is not a political document. The inclusion of the Influence Broker is not a statement about any political party, candidate, government, or ideology. It is a statement about a tool's properties.

The tool at issue is Gen AI — specifically, large language models and related architectures that generate text, images, and synthetic media from learned statistical distributions. That tool is being deployed across industrial control systems and information environments simultaneously. The same model family that a pipeline operator might use to recommend valve positions is being used by media organizations, political campaigns, and information operations to generate content at scale. The tool is the same. Its non-determinism is the same. Its incapacity for formal verification is the same. The functional requirement that a qualified human be interposed between its output and a consequential action is the same.

A framework that addressed the broker requirement in industrial control systems but not in the information environment would be analytically incomplete, not because the information environment is politically important, but because the same tool is operating in both environments

and presenting the same structural risk in both. Addressing both in a unified framework reflects analytical rigor, not political alignment.

Sector regulators reading this framework in the context of critical infrastructure compliance are not being asked to adopt, endorse, or engage with any position on democratic governance. The Command Broker requirements applicable to their sectors are fully self-contained in this document and in the applicable sector annexes. The Influence Broker material is provided for analytical completeness and because the qualification methodology — independent evaluation of AI output before consequential action, enforced by architecture rather than procedure — is shared. Regulators who wish to engage only with the Command Broker material may do so without loss of technical content.

5. Cross-Sector Architecture and Annex Structure

5.1 Relationship to the Sector Design Basis Series

Each sector in the DBA series establishes a Design Basis for the facilities within its scope. The Design Basis defines the envelope of credible events — including military action under the DBA-MA methodology — and the required protective architecture. Where that architecture includes a component that does not meet the determinism carve-out, the Command Broker requirement applies to it (FD-LD §4.3).

The sector Design Basis documents carry the Command Broker as an architectural requirement: the Bright Line must exist, and no above-line action of an in-scope component may execute without the Command Broker's authorization, discharged in the moment or, where FD-BL-D1 applies, in advance (FD-BL §4.5). The Design Basis does not include implementation details. Implementation detail — sector-specific consequence thresholds, band assignments, qualification standards, integration-pattern guidance, and audit criteria — is carried in the sector-specific implementation annexes to this framework.

5.2 Current and Anticipated Annexes

The following sector annexes are currently in development or anticipated:

- DBA-W Annex — Water Sector. Consequence thresholds: population-scale public health impact, loss of wastewater containment, loss of service to critical dependent facilities. Key operational context: SCADA-integrated AI, legacy system constraints, Oldsmar 2021 reference event. Qualification standard: water treatment and distribution system operations competency.
- DBA-GAS Annex — Natural Gas Transmission. Consequence thresholds: uncontrolled release events, electricity-gas interdependency cascade, cross-border supply disruption. Key operational context: compressor station automation, pipeline hydraulics, linepack

management. Qualification standard: gas transmission operations and pipeline control competency.

- DBA-OIL Annex — Petroleum Sector. Consequence thresholds: process excursion producing fire, explosion, or hazardous material release; refinery multi-product supply disruption. Key operational context: refinery distributed control systems, process unit interdependencies. Qualification standard: refinery process control and safety system competency.
- DBA-Nuclear Annex — Nuclear Facilities. Consequence thresholds: reactor protection system challenge, spent fuel thermal progression, and radiological release. Key operational context: defense-in-depth architecture, safety system independence requirements, DBA-MA military action stress conditions. Qualification standard: licensed reactor operator competency with AI-system evaluation training. Note: this annex exists as CB-Annex-Nuclear, held by Project 6 under this framework's conceptual parentage; its version state is recorded in CB-REG.
- Democratic Safeguards Annex — Information Environment. Consequence thresholds: content capable of distorting political outcomes at scale, synthetic media designed to undermine electoral processes, and AI-generated information operations targeting democratic institutions. Qualification standard: editorial judgment, source verification, and AI content evaluation competency. Developed under the Democratic Safeguards project.

5.3 Consequence-Derived Band Assignment

Each sector annex must assign its applications to the criticality bands through a consequence analysis specific to the sector’s operational context. The cross-sector framework establishes the methodology; the annexes provide the analysis. Band assignment is not transferable across sectors without independent analysis. The same action may fall below the Bright Line in one sector and above it in another, depending on the consequence environment. Band assignment is a property of the application and sets rigor; placement is a property of each action and is not derived from the band (§3.2a; FD-BL §4.3).

The consequence analysis scores the application in each of the six consequence categories it can reach — public health & safety, physical, environmental, economic & service-reliability, national security, and democratic & societal — nil in any it cannot. Within each category, severity is set by three modifiers: scale (facility, regional, national, societal), reversibility (recoverable within a defined timeframe vs. not), and conditions (normal, degraded, adversarial stress — the DBA-MA envelope). The application's band is the maximum severity reached in any single category; the category that sets it — the pinning category — is recorded, because it also identifies the governing regulatory regime. This is the aggregation rule defined in the Consequence-Severity Tier Model; the annexes supply the per-category thresholds.

6. Foundational References

Command Broker family instruments are cited in this document by designation, without version tokens. The current version of each is stated in CB-REG, which is the authoritative register for this family. The Foundational Definitions this framework cites as governing — FD-BL, the industrial Bright Line; FD-LD, which states scope; FD-BR, the Broker role-type; FD-BL-D1, mode of discharge; and FD-BL-D2, indication integrity — are cited the same way, and FD-REG is authoritative for their version state. This version was conformed against the October 2026 Edition of the Foundational Definitions (10.5281/zenodo.23117612).

Roxey, T.E. “Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers.” Version 29. Eclectic Technologies, 2026. [Primary source for the three-tier criticality framework and the Bright Line concept. This edition supersedes the prior ISAGCA-attributed “A Study of AI Risks in Critical Infrastructure — Vendor Agnostic,” Version 28.]

Roxey, T. DBA-W-Sector_Framework v1.12. Eclectic Technologies. [Water sector Design Basis establishing the Command Broker as a sector Design Basis requirement and developing the Section 5.5 command-broker architecture.]

Roxey, T. TAF-W_AI_QA_Water_Sector_Guide v1.6. Eclectic Technologies. [Water sector implementation guide for Command Broker architecture and deployment.]

Roxey, T. CB-CT Commissioning and Joint Training. Eclectic Technologies, 2026. [Companion module sequencing the vendor–AOO handoff for commissioning and joint training of the Command Broker.]

Democratic Safeguards Project. [In development. Source for the Influence Broker concept and the democratic governance application of the broker principle.]

Appendix A. Disposition of the v0.2 Conformance Markup

CB_Framework_Cross_Sector_v0_2_Conformance_Markup_v0_1 carries ten edits. Under CR-MD each is dispositioned by reading it against the current text of its target, edit by edit, and a markup whose edits have been read but not made is dispositioned as to its reading and not as to its target. This appendix records both readings.

Acted on before this version. Edit 2 is in: §3.2 now opens on the consequence criterion and states that the line is not the boundary between architectural layers. Edit 3 is in: §3.1 scopes the obligation to components within the determinism carve-out and §3.2a carries action-by-action routing.

Acted on at this version. Edit 5, the canonical term, is applied at all 13 occurrences in the body text, across eleven paragraphs including the §4 heading, §1, §4.1 to §4.3, §5.2 and §6. A fourteenth occurrence, in the v0.1 revision-history entry, is deliberately left as written under CR-VA §4.1: a record edited to match present state stops being evidence of what that version said. Edit 9 is completed for §3.3: the Mid and Lowest bands were left keyed to the band when the Highest band was conformed, and both are now conformed to placement, with the maloperation and availability entry routes distinguished. Edit 10 is taken in part: the family citation lead-in resolving to CB-REG is added at the head of §6 and the three stale pins in that section are repaired or stripped.

Read and not acted on; carried to v0.4. Edits 1, 4, 6, 7 and 8 — the ontology declaration at the head of §3, the reduction of §2 to a reference into FD-BR, the two-class Influence Broker definition under FD-SBL-D1, the correction of the claim that the two Brokers' parallel is precise, and the structural derivation of Influence Broker independence. They are deferred as a set because they conform this framework to definitions owned elsewhere, and because four of the five sit on a Line this framework's broker answers to but does not own. Projects 1, 6 and 8 were noticed of this deferral on 12 September 2026.

Read and not acted on; not yet scheduled. The remainder of Edit 9 at §5.3, which states the band-assignment method without restating the placement distinction, and the remainder of Edit 10 — the governing-definition citations and the cross-corpus scope note in §6.

One site is left as written deliberately. The §6 description of Generative AI Risk in Industrial Control Systems V29 calls it the primary source for the three-tier criticality framework. That is a description of a fixed published study, and a quotation is not a usage: it carries the wording of the document it describes and cannot conform ahead of it.

The markup therefore stays at the working root, beside its target, until the five carried edits are made.

Appendix B. Disposition of the Project 6 Comment of 12 September 2026

The three interface requirements at §3.7 were circulated as prose to Projects 6, 8 and 1 on 12 September 2026, and the circulation undertook that if the prose changed on comment it would be rebuilt from the comment rather than amended around it. Project 6 commented the same day, against the circulated prose alone and without opening any file of this project. The prose changed, and §3.7 was composed again from the comment; the pre-comment draft is a predecessor and was not a base for this version. This appendix records what was taken, what was decided rather than drafted, and what was held.

The comment found the prose sound in what it requires and incomplete in four ways, and asked for four sentences, none of which it supplied, with a fifth if wanted. All five are taken. The

integrity obligation is taken by attaching the three presented quantities to FD-BL-D2 and citing that instrument rather than restating its requirements, which is the propagation form its own conformance set directs. The adequacy criterion is taken by naming the three conditions the requirements compensate — cognitive overload, automation bias amplified by stress, and time compression — and by stating that each obligation is discharged only in a form that remains effective under all three. The fourth display condition is taken as the express rule rather than as a fourth state, which was the form the comment preferred and the one that states the governing point more exactly. The scope sentence is taken in the third of the forms the comment offered: the advance-discharge mode's equivalent obligations live in the sector annexes, and this section carries an express hook and an express requirement that each annex state them. The fifth is taken: the third requirement's plant-side precondition is acknowledged and its failure resolves to indeterminate, which by the section's own default is treated as above the line.

The characterization asked for is made. The three are no longer called duress requirements. They are compensating requirements written against three duress conditions, and their statement in this framework as unconditional standing obligations is recorded as a deliberate enlargement, so that a reader tracing them back through the DBA series gaps register or the nuclear application guide finds the trigger decided rather than dropped. The blanket disclaimer that these are obligations on the interface and not on the plant is narrowed to the first two requirements, which is where it is true, rather than being carried across a third requirement that breaches it.

One addition was not asked for and is made with them. The comment raised the absence of a stated failure posture as related and separable, and left to this project whether the cross-sector text carries an equivalent or leaves the posture to each annex. Silence resolves in the permissive direction at the moment of use, so this version states the floor — the above-line authorization is not available on an untrustworthy basis, and the placement resolves to indeterminate — and requires each annex to state the sector-conservative posture that sits above it.

Two items arising from the same comment are held and are not in this version. The gaps register entry that recommended assessing whether an inverter firmware precondition requires a dedicated section now reads onto this instrument, DBA-AI-CBR having been retired; the register is Project 6's to conform, the recommendation was passed here so that this project holds it before then, it was not asked for at this version and is not taken at it. And the observation about §11.4 of the Tiered Assessment Framework is Project 6's own to act on in its annex; no cascade is asserted here and none was asked. Nothing in the comment was rejected.

Appendix C. Conformance to the October 2026 Edition

This version conforms the framework to the October 2026 Edition of the Foundational Definitions, issued 2 October 2026 under version DOI 10.5281/zenodo.23117612. It answers Project 0’s note of 2 October 2026 on the governing scope text (Note_2026-10-02b, item 1), its note of the same date on the envelope author (Note_2026-10-02d), and the entry for this

framework at FD-BL §7. Every site was read against the issued masters, not against a note's paraphrase of them.

Changes of result. §3.1 no longer defines an in-scope component as generative or otherwise non-deterministic; it cites FD-LD §4.3, under which a deterministic component never formally tested is within scope. §2's statement that deterministic machine learning can operate autonomously is replaced by the carve-out's own guard. §5.1 no longer applies the requirement to "ML or Gen AI components" as a class, which reached deterministic components the carve-out removes and missed untested deterministic ones it does not. §3.2a no longer places the drafting of a maintenance work order below the line as a class; a work order that would take equipment out of service once acted upon is an action placed on that consequence (FD-BL §4.2). §3.7's carve-out rule now states both conditions of the property test, where v0.3 stated determinism alone.

Changes of mode. §3.1, §3.2, §3.5, §3.7 and §5.1 now state that the Command Broker discharges in the moment or, where FD-BL-D1 applies, in advance as the author of the pre-authorized envelope, and that a verified gatekeeper admits; v0.3 named pre-authorization as a bypass in §3.5 and left the envelope's author as "a qualified human" in §3.7. This is Decision C1 of 2 October 2026, cited at FD-BR §4.2 and FD-BL-D1 §7.

Wording and citation. FD-LD §3 and FD-BL §4.4 are replaced by FD-LD §4.3 wherever scope is cited. §3.7's integrity obligation now reaches inputs on which placement or discharge depends (FD-BL-D2 §4.1). §5.3's cross-sector sentence is restated for actions rather than applications, which completes the remainder of markup Edit 9. §6 takes the industrial half of markup Edit 10, the governing-definition citations; the influence half, FD-SBL and FD-SBL-D1, stays with Edits 6 to 8.

Carried to v0.5. Markup Edits 1, 4, 6, 7 and 8, for the reasons Appendix A gives; they remain with Projects 1, 6 and 8 on the notice of 12 September 2026. With them, and for the same reason, the sentences of §1, §4.1, §4.2 and §4.3 that derive the Influence Broker from the non-determinism of Gen AI: the Influence Broker's Line is FD-SBL's, which is Project 1's, and this version does not restate its scope.

Searched after the edits. The conclusion language of this project's instructions was searched across the body ("operate autonomously", "may operate autonomously", "above the line", "below the line", "requires a broker", "prohibited in", "by definition"), together with "Gen AI", "generative", "FD-LD §3", "pre-authoriz" and "envelope". Each remaining site was read in context. The surviving "Gen AI" sites are in §1's account of the two applications, §2's account of why the broker concept arose, §3.2's vendor-characterization and sophistication examples, and §4; none of them outside §4 states scope or placement for the Command Broker. §4.2's Command Broker half of the parallel does, by technology, and it is carried to v0.5 with the Influence Broker sentences because it is one sentence of a two-part parallel the deferred Edit 7 corrects. "Envelope of credible events" at §5.1 is the Design Basis sense, a set of events, and not

the operational sense FD-EV §4.3 renames; it is left as written. The sentence Project 6 raised on 3 October 2026, that neither the deploying organization nor the AI system holds accountability, does not occur in this framework; §2 places accountability with a named human decision-maker or governance body, and FD-BL §4.5 places it with the named Command Broker and the asset owner-operator.

Revision History

Version	Date	Description
0.1	May 2025	Initial draft. Establishes cross-sector broker concept, Command Broker and Influencer Broker definitions, three-tier criticality framework, qualification methodology, and annex structure. Addresses the relationship between the two broker types and the politicization question.
0.1 (housekeeping)	June 2026	Housekeeping correction. “Brightline” (one word) corrected to “Bright Line” (two words, title case) throughout all body text, section headings, and references, consistent with series-wide convention established in DBA-EN-SF §4.3. No substantive content changes.
0.2	Jul 2026	Reconciled to the Consequence-Severity Tier Model. §3.3 recast as criticality bands (Highest/Mid/Lowest) on the six-category take-the-max basis; naming crosswalk demoted to a legacy reading aid and the type-words (safety-critical/quality-of-service/administrative) withdrawn as band definitions and returned to the error classes; §5.3 promoted to the six-category consequence basis with the pinning-category rule; band vocabulary applied to the framework's own normative text. Descriptions of the fixed source study (V29) left verbatim.
0.2 (cite refresh)	Jul 2026	Citations refreshed to current corpus versions: CB-Framework v0.2, Command Broker Implementation Guide V0.2, CB-CT v0.2, DBA-VC v0.8. Mechanical version-reference pass; no substantive content change. (Held pending decision: the NERC/BES-Variant TAF reference and the ISA Parent Framework V1.2 label.)
0.3	Sep 2026	Issued 12 September 2026 from Project 2, the family's governance layer having moved here on the author's determination of 11 September 2026; the first revision of the family parent issued under that ownership, and the first firing of the obligation that such a revision be noticed to the projects holding its applications. The three interface requirements at §3.7 were circulated as prose to Projects 6, 8 and 1 on 12 September 2026 under the order agreed at the D-1 answer of 4 September 2026. Project 6's comment of the same date found the prose sound in what it requires and incomplete in four ways, and §3.7 was rebuilt from that comment rather than amended around it, as the circulation undertook; the pre-comment draft is archived as a predecessor and was not a base for this version. New §3.7 states the Command Broker interface obligations — calibrated confidence with its basis and uncertainty, consequence preview, and continuous placement-state display — written here

Version	Date	Description
		rather than moved. It names the three conditions those requirements compensate and states the criterion of adequacy that follows from them, recording as deliberate their statement as unconditional standing obligations; attaches the three presented quantities to FD-BL-D2 by citation and states the failure posture that follows, an integrity status that cannot be established being displayed rather than omitted; rules that an action carved out of the obligation by FD-LD is never rendered as below the line, the carve-out being scope and not placement, which is the correction Tiered Assessment Framework §2.4 carries for the scope of the obligation itself; keys the display to the action and not to the component; acknowledges the third requirement's plant-side precondition and resolves its failure to indeterminate; and states which mode the section governs, hooking the advance-discharge mode of FD-BL-D1 to the sector annexes. The three are no longer called duress requirements. Appendix B records the disposition of the comment. Elsewhere in this version: “Influencer Broker” conformed to the canonical “Influence Broker” at all thirteen body occurrences (FD-BR §6.1), the fourteenth, in the v0.1 revision-history entry, left as written under CR-VA §4.1. §4.2’s residual layer-boundary formulation corrected to the criterion §3.2 already carries. §3.3’s Mid and Lowest bands conformed to placement, with the maloperation and availability entry routes distinguished; the Highest band was already conformed and is unchanged. §5.2 corrected: the nuclear annex exists and is Project 6’s. §6 takes the CB-REG citation lead-in; DBA-W-Sector_Framework repaired v1.0 → v1.12 and TAF-W v1.2 → v1.6 against Project 9’s masters, and the CB-CT token stripped. Appendix A records the disposition of all ten markup edits. Five edits are carried to v0.4.
0.4	Oct 2026	Issued 3 October 2026, conforming the framework to the October 2026 Edition of the Foundational Definitions (10.5281/zenodo.23117612) under Project 0’s work order of that date. Scope is cited to FD-LD §4.3 rather than defined: §3.1 no longer defines an in-scope component as generative or otherwise non-deterministic, §2 no longer says deterministic machine learning can operate autonomously, and §5.1 no longer applies the requirement to ML or Gen AI as a class. §3.2a no longer places the drafting of a maintenance work order below the line as a class (FD-BL §4.2, §4.3). The Command Broker discharges in the moment or in advance as the author of the pre-authorized envelope, and the verified gatekeeper admits (Decision C1; FD-BR §4.2; FD-BL-D1 §7), at §3.1, §3.2, §3.5, §3.7 and §5.1; §3.5 no longer names pre-authorization as a bypass without qualification. §3.7’s carve-out rule states both conditions of the property test, and its integrity obligation reaches placement and discharge inputs (FD-BL-D2 §4.1). §5.3 completes markup Edit 9; §6 takes the industrial half of Edit 10. Edits 1, 4, 6, 7 and 8 and the Influence Broker’s scope sentences are carried to v0.5. Appendix C records the pass. Republished to the publications site at the same path under CR-PB §5.
0.5	Nov 2026	Issued 11 November 2026 on the author’s instruction of that day. One sentence of the legacy naming crosswalk at §3.3 is corrected: the type-words once

Version	Date	Description
		attached to the numbered bands no longer “name error classes”, a term not adopted under the vocabulary determination of 7 September 2026, but echo the three tiers of the error axis, which the Tiered Assessment Framework defines as safety-critical, quality-of-service and compliance-driven. The legacy list keeps “administrative” as a quotation of the older texts. Nothing else changes. Noticed to Projects 6, 8 and 1 before issue, in Notice_2026-11-11e. The three items held for a later revision (an inverter-firmware section, the reach of the nuclear hardening review to §3.7, and failing to a stop as a Command Broker property) are not taken here. The published v0.4 is left standing as a dated draft under the CR-PB interim default.